

Data Science

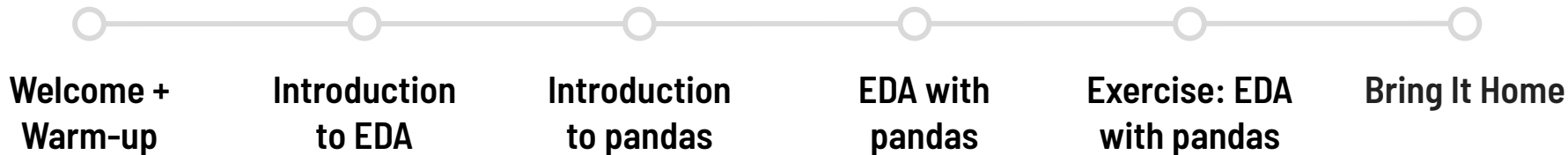
Introduction to Exploratory Data Analysis (EDA) and pandas

Exploratory Data Analysis



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Explain the difference between EDA for exploration versus EDA for modeling.

2

Define pandas Series and DataFrame types and their uses.

3

Understand the benefit of vectorized dataset operations.

4

Read in a dataset with the pandas library.

5

Explore the characteristics of a dataset at a high level with pandas.

Introduction to EDA

Reasons To Explore Data

EDA for exploration

To understand what's in a data set, investigate missing values, summary statistics etc.

EDA for modeling

To find relationships in the data, understand what correlates with our target, and create new features for use in a machine learning model

Typical EDA tasks

EDA for exploration

- Look at the data at a glance.
- Identify continuous vs. categorical data.
- Investigate missing values.
- Explore summary statistics.
- Visualize distributions in the data.

EDA for modeling

- Identify and visualize relationships in the data (i.e. correlation).
- Find features that correlate with our target.
- Create new features by transforming existing columns.

Introduction to pandas

What is pandas?

1

pandas is a Python library for exploring, manipulating, joining, and even visualizing data.

2

pandas is **fast**: faster than manipulating data with “raw” Python!

3

pandas helps make your analysis fast and reproducible.

pandas Terminology

pandas introduces new data types: `Series` and `DataFrame`

- A `Series` is one-dimensional - a single row or column
- A `DataFrame` is a two-dimensional “table” of data
 - In a `DataFrame`, each row and column is a `Series`



pandas Terminology

Each column is a
Series

Each row is
a Series

Customer ID	Customer Forename	Customer Surname	Customer Age
505	John	Smith	55
196	Templeton Arthur	Peck	43
947	H. M.	Murdock	36
952	Bosco Albert	Baracus	31

The entire data set is a DataFrame

Why pandas?

- pandas is the most popular data analysis framework for Python
- pandas is **fast**, because:
 - It uses numpy arrays under the hood, which make use of C/Cython
 - It uses **vectorized** operations
- Vectorized operations make it possible to manipulate entire columns at once (without for loops!)



Introduction to pandas

Let's see it for ourselves!

- Open **01-intro-to-pandas.ipynb**



Break

time



EDA with pandas

EDA with pandas

Let's see it for ourselves!

- Open **02-eda-with-pandas.ipynb**



Exercise: EDA with pandas

Exercise: EDA with pandas

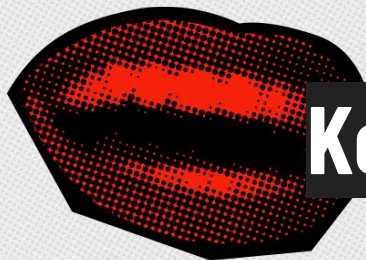


Your turn!

- Time to do some further analysis of the data and answer a stakeholder question.
- open **03-exercise-eda-with-pandas.ipynb**



Bring It Home



Key takeaways

Introduction to EDA with pandas

1

EDA for Exploration

Understand the basics of your data

Look at data types, manipulate columns, calculate summary statistics

2

EDA for modeling

Investigate relationships in the data.

Create new features.

Look for features to use in a machine learning model.

3

What is pandas?

Data analysis library in Python.

Introduces Series and DataFrame types.

Can be used for exploring, cleaning, and visualizing data.

4

Why use pandas?

It's fast.

Supports many types of data source.

Wide adoption in data science.

DON'T FORGET:

Exit tickets



